

A Comparative Analysis of MLR and Linear SVM using Wine Dataset

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Abstract— This study presents a comparative analysis of Multinomial Logistic Regression (MLR) and Linear Support Vector Machine (SVM) for multiclass classification using the Wine dataset. The dataset consists of chemical features used to classify wine cultivars into three categories. Both models were implemented using scikit-learn pipelines with data standardization and evaluated through 10-fold cross-validation. Hyperparameter tuning was performed using GridSearchCV to determine the optimal configurations, particularly focusing on the regularization parameter C . Results show that both models achieved the same highest mean cross-validation accuracy of 0.985714. However, in final testing, Linear SVM outperformed MLR by achieving perfect accuracy, Kappa, and F1-score, while MLR also demonstrated strong and consistent performance. Feature importance analysis revealed that both models identified similar key attributes, such as color intensity and proline, as significant predictors. Overall, the findings indicate that while both probabilistic and geometric models are effective, the tuned Linear SVM provides slightly better classification performance for this dataset.

Keywords—*Multinomial Logistic Regression, Linear Support Vector Machine, Wine Dataset, Hyperparameter Tuning, Feature Importance*

I. INTRODUCTION

Multiclass classification is a fundamental machine learning task that involves assigning input data into one of several predefined categories based on observed features. Among the commonly used approaches for this task are Multinomial Logistic Regression (MLR) and Support Vector Machines (SVM), both of which have demonstrated strong performance across various domains due to their solid theoretical foundations and practical effectiveness. Multinomial Logistic Regression is a probabilistic, parametric model that utilizes the softmax function to estimate class probabilities, making it suitable for problems requiring interpretable outputs. In contrast, the Linear Support Vector Machine is a geometric, margin-based algorithm that identifies an optimal hyperplane to separate classes by maximizing the margin between them, which often leads to robust generalization performance [1].

Despite their differences in learning principles, both models remain widely applied in modern machine learning applications such as bioinformatics, chemical analysis, and pattern recognition [2]. In this study, the Wine dataset, originally introduced as a benchmark dataset for

classification problems, is utilized to evaluate and compare the performance of these two models. The dataset consists of chemical properties derived from different wine cultivars, making it suitable for analyzing classification techniques in chemical profiling tasks. The objective of this activity is to implement, fine-tune, and compare Multinomial Logistic Regression and Linear Support Vector Machine using standardized data and cross-validation. Furthermore, the study aims to analyze how the regularization parameter (C) influences model performance in terms of accuracy and other evaluation metrics, and to determine which approach provides better classification performance for this dataset [3].

II. METHODOLOGY

This section presents the organization and logical structure of the implemented program. The workflow is divided into several key components, including setup and imports, data acquisition, data preprocessing, model development, hyperparameter tuning, and evaluation of results.

A. Setup & Imports

This part of the program is responsible for importing the necessary libraries required for data processing, modeling, and evaluation. The scikit-learn library is primarily used for loading the dataset, building machine learning models, and performing hyperparameter tuning. Additionally, matplotlib.pyplot is utilized for data visualization, particularly for generating confusion matrices and graphical comparisons of results. These libraries provide the essential tools needed to efficiently implement and evaluate the classification models.

B. Data Acquisition

The dataset used in this study is the Wine dataset obtained from `sklearn.datasets.load_wine()`. This dataset contains 13 chemical attributes of wine samples along with their corresponding class labels, representing three different cultivars. The dataset is loaded into memory and structured into input features (X) and target labels (y). The input features represent the chemical composition of each sample, while the target labels indicate the class category. This separation allows the model to learn patterns from the features and use them to predict the correct class labels.

C. Data Preprocessing

Before training the models, the dataset is divided into training and testing subsets using an 80/20 split. The training set is used for model learning and hyperparameter tuning, while the testing set is reserved for final evaluation. To ensure consistent scaling of features, `StandardScaler()` is applied to normalize the data. This step is particularly important for algorithms such as Support Vector Machines, which are sensitive to feature magnitudes. A Pipeline is implemented to combine scaling and model training into a single process, preventing data leakage and ensuring proper application of transformations.

DATA SPLIT INFORMATION

	Dataset	Instances	Percentage
0	Training Set	142	80%
1	Testing Set	36	20%

Fig. 1. Data Split Information

D. Model Development

Two classification models are implemented in this study: Multinomial Logistic Regression and Linear Support Vector Machine. The Multinomial Logistic Regression model uses the softmax function to estimate class probabilities and is configured with the lbfgs solver. On the other hand, the Linear SVM model, implemented using `LinearSVC`, focuses on finding the optimal hyperplane that maximizes the margin between classes. Both models are integrated within pipelines to streamline preprocessing and training.

E. Hyperparameter Tuning

To enhance model performance, hyperparameter tuning was performed using `GridSearchCV` with 10-fold cross-validation. For the Multinomial Logistic Regression model, multiple configurations were explored by varying the regularization parameter C with values of 0.001, 0.01, 0.50, 1, 10, and 100. In addition, different combinations of solvers and penalties were tested, including the lbfgs solver with l2 penalty and the saga solver with l1 penalty. The effect of class weighting was also examined by comparing None and balanced. These different experimentation allowed the identification of the most optimal configuration based on cross-validation performance.

Similarly, the Linear Support Vector Machine model was tuned by testing different values of the regularization parameter C over the same range. Additional hyperparameters were explored, including penalty types (l1 and l2), loss functions (hinge and squared_hinge), and the dual formulation (True or False). The inclusion of class weights (None and balanced) was also evaluated to determine its effect on handling class distribution. By testing multiple parameter combinations, all of this was done to determine whether additional hyperparameter adjustments could further improve the model's classification performance. Furthermore, the best combination was selected based on the highest mean cross-validation accuracy

III. EXPERIMENTAL DESIGN

This section presents the experimental setup and analysis of how different hyperparameters affect the performance of the Multinomial Logistic Regression (MLR) and Linear Support Vector Machine (SVM) models.

A. Multinomial Logistic Regression (MLR)

The model was evaluated using different combinations of solvers, penalties, class weights, and regularization parameter values. The result shows that the model's performance was highly influenced by the choice of the regularization parameter C . Lower values of C , such as 0.001 and 0.01, resulted in lower accuracy due to strong regularization, which constrained the model and prevented it from capturing important patterns in the data, leading to underfitting. As the value of C increased, the model achieved higher cross-validation accuracy, indicating better fitting of the data. For the lbfgs solver with l2 penalty, the model reached a stable accuracy of 0.978571 at $C=0.5$, 1, and 10, showing consistent performance across these values. However, when C was increased to 100, the accuracy slightly decreased, suggesting that increasing model complexity beyond a certain point does not improve generalization.

For the saga solver with l1 penalty, both `class_weight=None` and `class_weight='balanced'` showed similar behavior. The model achieved a higher accuracy of 0.985714 starting at $C=0.5$, which remained consistent for larger values of C . This indicates that the model reached its optimal performance at moderate values of C . It was also observed that both class weight settings produced the same accuracy. However, the model selected `class_weight=None` as the best configuration because `GridSearchCV` chooses the first occurrence of the best-performing parameter combination when multiple configurations achieve identical scores. Additionally, since the dataset is already balanced, applying class weighting does not significantly affect the model's performance. Overall, the saga solver with l1 penalty provided the best performance, while the lbfgs solver with l2 penalty produced stable but slightly lower accuracy.

B. Linear Support Vector Machine (Linear SVM)

The Linear Support Vector Machine model was evaluated using different combinations of penalty types, loss functions, dual settings, class weights, and regularization parameter values. The results showed that the model's performance was highly dependent on the choice of the regularization parameter C and the selected loss function. From a geometric perspective, SVM works by finding a hyperplane that maximizes the margin between different classes. The parameter C directly controls this margin, where a smaller value of C allows a wider margin but may result in more misclassifications, while a larger value of C produces a narrower margin with fewer misclassifications.

Based on the results, very small values of C , such as 0.001, resulted in lower accuracy, indicating underfitting due to excessive regularization. As the value of C increased to 0.01 and above, the model showed a significant improvement in performance. For the l1 penalty with squared_hinge loss and dual=False, the model achieved a

high accuracy of 0.985714 starting from $C=0.5$ and maintained this performance for larger values of C .

Similarly, for the l2l penalty with hinge loss and dual=True using balanced class weights, the model showed gradual improvement as C increased, reaching 0.978571 at $C=1, 10$, and 100 . For the l2 penalty with squared_hinge loss and dual=False, the model achieved consistent high performance of 0.985714 from $C=0.5$ onwards, indicating that this configuration provided more stable and reliable results compared to others.

Overall, the results indicate that the squared_hinge loss function generally produced more stable and higher accuracy compared to the standard hinge loss. The choice of penalty and dual formulation also affected performance, with l2 penalty and dual=False providing the most consistent results. Similar to the MLR model, class weighting had minimal impact on performance due to the balanced nature of the dataset. These findings show that while Linear SVM can achieve high accuracy, its performance is more sensitive to hyperparameter selection, particularly the value of C and the choice of loss parameter.

IV. COMPARATIVE RESULTS

This section presents the comparison of the best-performing Multinomial Logistic Regression and Linear Support Vector Machine models based on hyperparameter optimization.

A. Hyperparameter Optimization

BEST MODEL SUMMARY

	Metric	Value
0	Best C	0.5
1	Best Solver	saga
2	Best Penalty	l1
3	Best Class Weight	None
4	Best Mean CV Accuracy	0.985714
5	CV Standard Deviation	0.042857

BEST MODEL SUMMARY (SVM)

	Metric	Value
0	Best C	0.01
1	Best Penalty	l2
2	Best Loss	squared_hinge
3	Best Dual	False
4	Best Class Weight	None
5	Best Mean CV Accuracy	0.985714
6	CV Standard Deviation	0.028571

Fig. 2. Best Model of MLR and SVM

Based on the hyperparameter tuning results, both models achieved the same highest mean cross-validation accuracy

of 0.985714. For Multinomial Logistic Regression, the best configuration used $C=0.5$, $C = 0.5$, the saga solver, l1l1l1 penalty, and class_weight=None. For Linear SVM, the best configuration used $C=0.01$, $C = 0.01$, l2l2l2 penalty, squared_hinge loss, dual=False, and class_weight=None. Although both models reached the same mean accuracy, MLR achieved this performance using a moderate CCC value, while SVM achieved it using a smaller CCC value, showing that SVM performed better with stronger regularization.

B. Final Test Metrics

SUMMARY TABLE

	Model	Accuracy	Kappa	Weighted F1
0	MLR	0.972222	0.957746	0.97197

SUMMARY TABLE (SVM)

	Model	Accuracy	Kappa	F1
0	Linear SVM	1.0	1.0	1.0

Fig. 3. Comparison of Accuracy, Kappa, and F1 of MLR and SVM

Based on the results, Multinomial Logistic Regression achieved an accuracy of 0.972222, a Cohen's Kappa score of 0.957746, and a weighted F1-score of 0.97197, indicating strong and consistent classification performance across all classes.

In comparison, the Linear Support Vector Machine model achieved perfect scores, with an accuracy of 1.0, a Kappa score of 1.0, and an F1-score of 1.0. This indicates that the model was able to correctly classify all test samples without any misclassification.

Overall, both models performed well on the test dataset, but Linear SVM slightly outperformed Multinomial Logistic Regression in terms of final evaluation metrics. The results suggest that while MLR provides stable and reliable performance, SVM has the ability to achieve perfect classification.

C. Importance Comparison

The comparison of feature importance between Multinomial Logistic Regression (MLR) and Linear Support Vector Machine (SVM) based on the mean absolute values of their coefficients. Based on the results, MLR generally assigns higher importance values across most features compared to SVM, indicating that it distributes influence more broadly among the chemical attributes. In particular, features such as color_intensity, proline, flavanoids, alcohol, and alkalinity_of_ash show high importance in MLR, suggesting that these variables play a major role in distinguishing between wine classes.

In contrast, the Linear SVM model assigns lower overall importance values but still highlights similar key features, such as proline, color_intensity, and alcohol, as influential in classification. This indicates that both models agree on the most relevant features, even though the magnitude of their contributions differs. Additionally, SVM tends to concentrate importance on fewer features, reflecting its goal

of maximizing class separation using the most critical variables.

Overall, the comparison shows that while MLR provides a more distributed interpretation of feature importance, SVM focuses on the most dominant features. This difference reflects the probabilistic nature of MLR and the geometric approach of SVM, where MLR considers the contribution of all features, while SVM emphasizes those that are most important for defining the decision boundary.

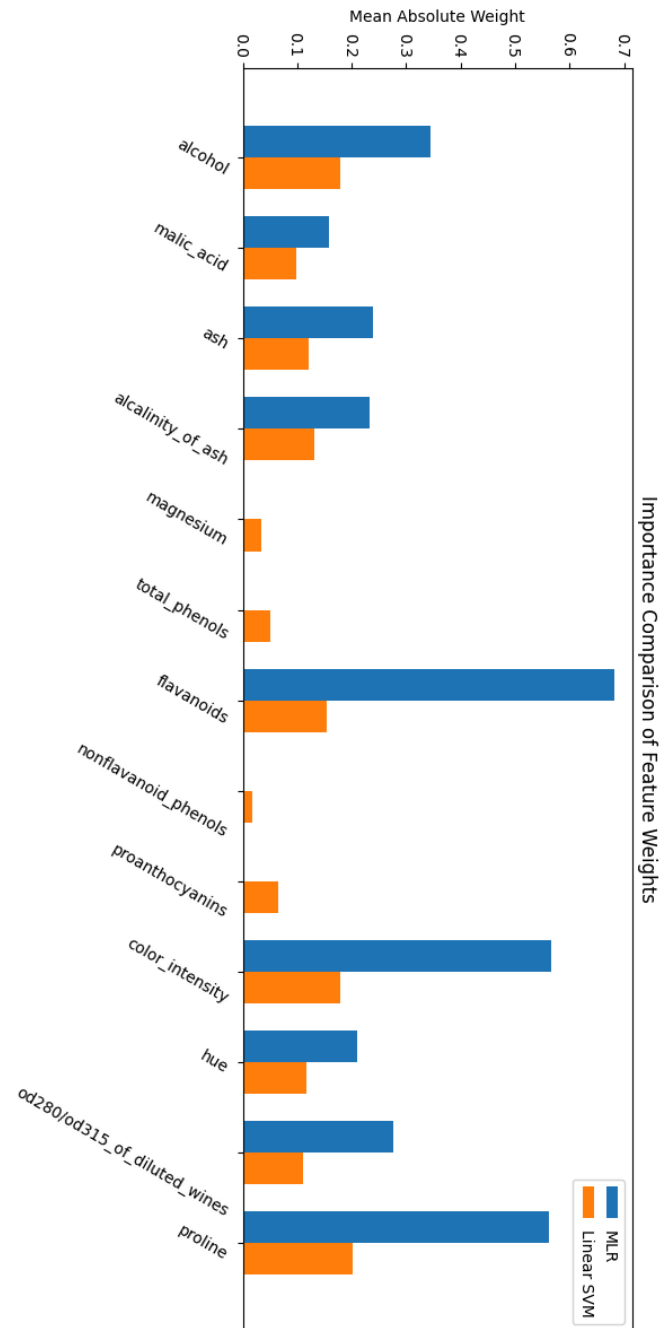


Fig. 4. Feature Importance

V. CONCLUSION

This study used Multinomial Logistic Regression and Linear Support Vector Machine to classify wine cultivars based on chemical features. The analysis showed that both models were effective, with both achieving the same highest mean cross-validation accuracy of 0.985714 during hyperparameter tuning. For MLR, the best model used $C=0.5$, the saga solver, l1 penalty, and no class weighting. For Linear SVM, the best model used $C=0.01$, l2 penalty, squared_hinge loss, dual=False, and no class weighting.

Among the two models, Linear SVM produced the best final testing performance. It achieved an accuracy, Kappa, and F1-score of 1.0, meaning it correctly classified all test samples. Meanwhile, MLR also performed strongly, with an accuracy of 0.972222, Kappa of 0.957746, and weighted F1-score of 0.97197. These results suggest that although both models are reliable, the geometric approach of Linear SVM was slightly more effective for this dataset.

The feature importance comparison also showed that both models identified similar important chemical attributes, especially color_intensity, proline, flavanoids, and alcohol. These features contributed strongly to the classification of wine cultivars. Overall, the results confirm that both probabilistic and geometric models can perform well on chemical classification problems, but the tuned Linear SVM provided the strongest final result in this experiment.

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